Catalogue of AI Tests and Controls

Version 0.3 – March 2026

The Catalogue indexes testing solutions and control checklists designed to evaluate the trustworthiness and regulatory compliance of AI systems. It is open to contributions and encompasses:

- A **public website** where anyone can browse, explore, and submit solutions.
- A **downloadable version** from the Sandbox Configurator's GitHub repository that runs locally in a browser, enabling one-click plugin installation directly into the host environment.

Each solution in the Catalogue is integrated into the Sandbox Configurator through a **plugin,** i.e. a self-contained adapter that encapsulates the solution's execution logic and maps its outputs to the system's shared data model. Solutions listed in the Catalogue that have not yet been implemented as plugins serve as open contribution requests, inviting the community to develop them.

The catalogue will also include datasets in future releases; however, the current release focuses exclusively on plugins.

1. Adding a Plugin

To contribute a plugin implementation, contributors must:

- Write metadata describing the solution, including tags to aid discoverability.
- Map the solution's outputs to the project's shared data model, i.e. the harmonisation layer used across the Sandbox Configurator to represent test and control check results in a consistent, DBMS-agnostic format.
- Implement the required plugin hooks in accordance with the Plug-in documentation in this Github repository: https://github.com/lux-ai-factory/rfc.

All submissions undergo a vetting process overseen by project administrators before becoming available for installation. Providers may additionally declare which regulatory requirements their solution addresses.

As providers may naturally tend towards broader claims, one avenue worth exploring would be to involve a National Competent Authority or the AI Office in validating declared coverage, though the feasibility of such an approach remains to be assessed.

2. Discovering a Plugin

Users can filter plugins by category using tag, as shown in Figure 1.

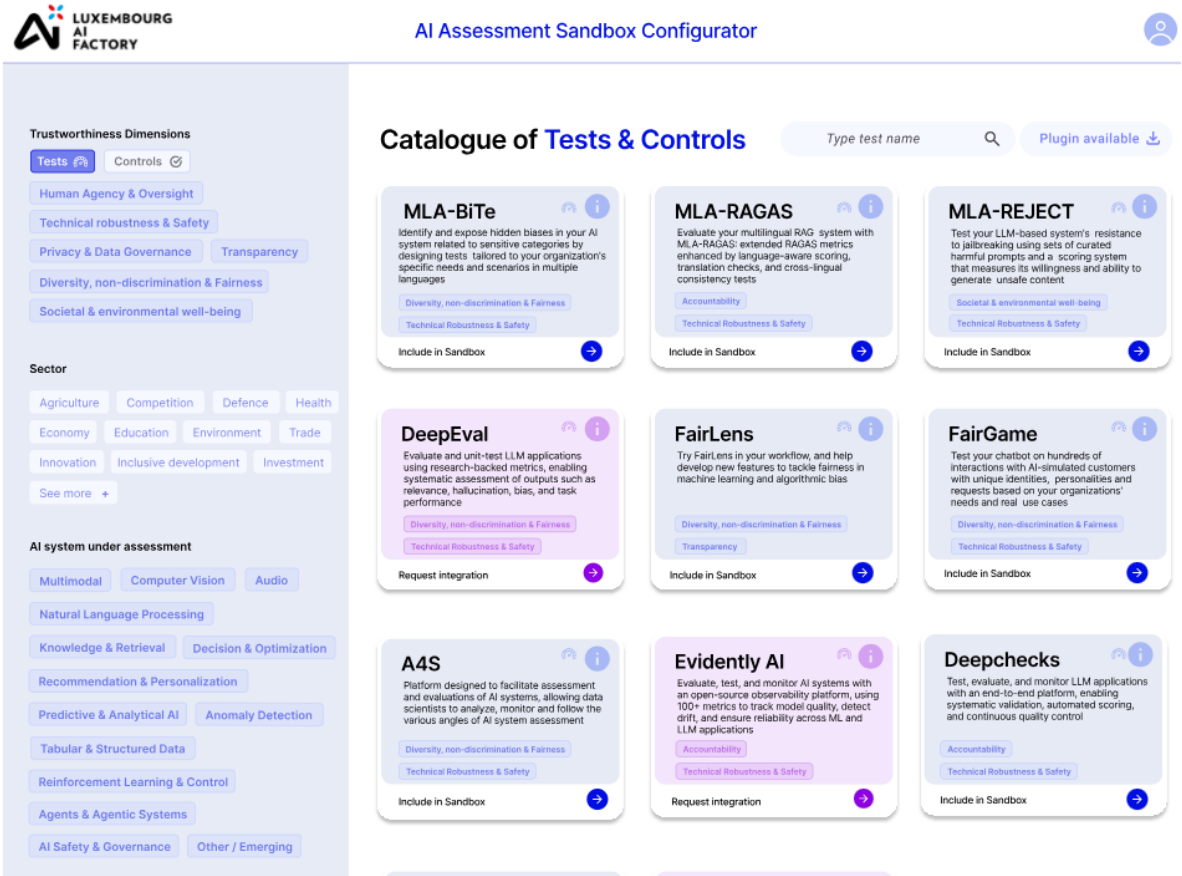

Figure 1. User interface of the catalogue. Users can explore plugins using the tags on the right that serve as filters.

3. Plugin metadata

The following metadata template is used to describe and register a new plugin in the catalogue.

Field	Format	Explanation
Name	*Text*	Commercial name of plugin.
Description	*Text*	Brief overview of the testing solution, including its purpose, scope, and main functionalities.
License	[TAG]	Type of license governing the use and distribution of the solution.
Verification type	Technical test/control	Specifies the type of verification activity performed.
Provider	*Text*	Name of the organisation, team, or entity responsible for developing or maintaining the solution.
Project	Hyperlink	Link to the associated project page, repository, or official documentation.
Branch	*Text*	Development branch or implementation stream.
Version	*Text*	Version number of the testing solution.
Project maturity	*Text*	Current maturity level (e.g., concept, prototype, validated in lab, deployed, maintained).
[OPTIONAL] Scientific reference	*APA Text*	Reference to related scientific publication(s) in APA format, if applicable.
Verification targets	[TAGS]	Keywords categorising the solution's purpose. Multiple selections allowed.
Sector	[TAGS]	Application domain(s) where the solution is relevant. Multiple selections allowed.
Target systems	[TAGS]	Types of AI systems that the solution can verify. Multiple selections allowed.
[OPTIONAL] Target legal requirements	*Text*	Textual description of

4. Verification targets – Testing [TAGS]

Keywords categorising the purpose of plugins for technical testing. This taxonomy draws largely from the joint paper by LIST and RISE, complemented by the OWASP Top 10 for Large Language Model Applications with respect to technical robustness and safety.

Requirement	Sub-dimension
Human Agency and Oversight	Override & opt-out mechanism testing
	Audit trail & logging completeness
Technical Robustness and Safety	Stress testing
	Adversarial input testing
	Out-of-distribution testing
	Drift detection
	Model calibration
	Prompt Injection
	Insecure Output Handling
	Training Data Poisoning
	Sensitive Information Disclosure
	Insecure Plugin Design
	Excessive Agency
	Overreliance
	Model Theft
Privacy and Data Governance	PII detection testing
	Data anonymisation / re-identification risk
	Data retention policy violation detection
Transparency	Local interpretability
	Global interpretability
	Explanation consistency testing
Diversity, Non-discrimination and Fairness	Bias detection on data
	Bias detection on system
Societal and Environmental Wellbeing	Energy consumption tracking
	Hardware utilisation profiling

5. Verification targets – Controls [TAGS]

Keywords categorising the purpose of solutions for controls checking. This taxonomy draws largely from the joint paper by LIST and RISE.

Requirement	Sub-dimension
Human Agency and Oversight	Supervisory control
	Strategic governance
	Informed consent
	User preferences
	Audit trails
Technical Robustness and Safety	Fail-over architecture
	Human fallback
Privacy and Data Governance	Purpose limitation
	Scope restriction
	Pseudonymisation
	Role-based access control
	Lineage tracking
	Metadata versioning
	Consent tracking
	Granular revocation
Transparency	Model cards
	Datasheets
	AI identity notices
	Disclaimers
	Risk communication
	Understandable language
Diversity, Non-discrimination and Fairness	Sampling strategies
	Accessibility
	Participatory design
Societal and Environmental Well-being	Rights-based system
	Independent ethics committee
Accountability	RASCI charts
	Logging
	External reporting interfaces
	Redress systems
	Version-controlled records
Quality Management	Quality Policy Declaration
	Control of documented information
	Internal Auditing
	Continuous Improvement
Risk Management	Description of AI system
	Structured risk identification

	Core classification step
	Finalise classification decision
	Outline potential risk areas
	Establishing PMM plan
	Perform CAPA
	Compliant incident reporting
Technical Documentation	System overview and intended purpose
	Architecture and design specifications
	Dataset provenance
	Structured compliance mapping
Record-keeping	Operational logging system
	Version control
	Full lineage
	Retention/deletion policies

6. Target systems

Types of AI systems that the solution can verify. This taxonomy is largely based on Hugging Face's tags. Since Hugging Face categorises AI models rather than AI systems, adaptations were necessary, including the addition of categories not covered in the original, such as multi-agent systems.

Category	Sub-category / Capability
Multimodal	Multimodal Understanding
	Multimodal Generation
	Multimodal Retrieval
Computer Vision	Image Understanding & Classification
	Object Detection & Segmentation
	Image Generation & Editing
	Video Understanding & Generation
	3D Understanding & Generation
	Depth & Spatial Estimation
Natural Language Processing	Text Understanding & Classification
	Question Answering
	Text Generation & Summarization
	Sentence & Semantic Similarity
	Information Extraction
Audio	Speech & Audio Recognition
	Speech & Audio Generation
Knowledge & Retrieval	Retrieval-Augmented Generation (RAG)
	Semantic Search
	Knowledge Graph Systems
	Document Intelligence
Recommendation & Personalization	Collaborative Filtering
	Content-Based Recommendation

	Hybrid & Sequential Recommendation
	Personalization
Predictive & Analytical AI	Predictive Analytics
	Time Series Forecasting
	Risk Scoring & Assessment
	Causal Inference
Anomaly Detection	Anomaly & Outlier Detection
	Monitoring & Alerting
Tabular & Structured Data	Tabular Classification & Regression
	Feature Engineering & Selection
Decision & Optimization	Decision Support Systems
	Constraint Optimization
	Simulation & Scenario Modeling
Reinforcement Learning & Control	RL & Sequential Decision-Making
	Robotics & Embodied AI
	Process Control & Automation
Agents & Agentic Systems	Task Planning & Decomposition
	Tool Use & API Orchestration
	Memory & Context Management
	Multi-Agent Coordination
	Human-in-the-Loop Systems
	Autonomous Workflow Execution
AI Safety & Governance	Content Moderation
	Privacy-Preserving AI
Other	Graph Machine Learning
	Neuromorphic & Edge AI
	Quantum-Classical Hybrid AI
	Synthetic Data Generation

7. Sector

Application domain(s) where the solution is relevant. This taxonomy is sourced from the OECD catalogue of AI tools.

Sector
Agriculture
Competition
Corporate governance
Defence
Digital Economy
Economy
Education
Employment & labour
Environment

Finance and insurance
Health
Inclusive development
Industry & entrepreneurship
Innovation
Investment
Other
Public governance
Public sector
Science & technology
Social & welfare issues
Tax Transport
Trade

8. License

Type of license governing the use and distribution of the solution. The list of licenses is sourced from Hugging Face.

License	Open / Closed
apache-2.0	Open
mit	Open
openrail	Open
cc-by-4.0	Open
afl-3.0	Open
cc-by-sa-4.0	Open
gpl-3.0	Open
artistic-2.0	Open
bsd-3-clause	Open
agpl-3.0	Open
cc0-1.0	Open
wtfpl	Open
unlicense	Open
bsl-1.0	Open
bsd	Open
bsd-2-clause	Open
gpl	Open
cc-by-sa-3.0	Open
bsd-3-clause-clear	Open
cc-by-2.0	Open
cc-by-3.0	Open
cdla-permissive-2.0	Open
gpl-2.0	Open
cc-by-2.5	Open

lgpl-3.0	Open
osl-3.0	Open
mpl-2.0	Open
gfdl	Open
pddl	Open
ecl-2.0	Open
ms-pl	Open
etalab-2.0	Open
epl-2.0	Open
cdla-sharing-1.0	Open
odc-by	Open
cdla-permissive-1.0	Open
lgpl	Open
eupl-1.2	Open
eupl-1.1	Open
odbl	Open
lppl-1.3c	Open
zlib	Open
epl-1.0	Open
lgpl-2.1	Open
isc	Open
ncsa	Open
lgpl-lr	Open
postgresql	Open
ofl-1.1	Open
bigscience-openrail-m	Open
creativeml-openrail-m	Closed
cc-by-nc-4.0	Closed
llama3	Closed
gemma	Closed
llama2	Closed
openrail++	Closed
cc-by-nc-sa-4.0	Closed
bigscience-bloom-rail-1.0	Closed
bigcode-openrail-m	Closed
llama3.3	Closed
cc	Closed
cc-by-nc-nd-4.0	Closed
cc-by-nc-2.0	Closed
cc-by-nd-4.0	Closed
cc-by-nc-3.0	Closed
apple-amlr	Closed
cc-by-nc-nd-3.0	Closed
cc-by-nc-sa-2.0	Closed

cc-by-nc-sa-3.0	Closed
apple-ascl	Closed
deepfloyd-if-license	Closed
fair-noncommercial-research-license	Closed
c-uda	Closed
llama4	Closed
grok2-community	Closed
openmdw-1.0	Closed
intel-research	Closed
h-research	Closed
open-mdw	Closed
Other	NA